

AIM : RSA Encryption and Decryption

Implement the RSA algorithm for public-key encryption and decryption, and explore its properties and security considerations.

CLI Version :

Code :

```
from math import gcd

# Function to check if a number is prime

def is_prime(n):
    if n < 2:
        return False
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False
    return True

# Function to find modular inverse
def mod_inverse(e, phi):
    for d in range(1, phi):
        if (d * e) % phi == 1:
            return d
    return None

# User Input

p = int(input("Enter first prime number (p): "))
q = int(input("Enter second prime number (q): "))

# Check if inputs are prime

if not is_prime(p) or not is_prime(q):
    print("Error: Both numbers must be prime.")
    exit()

# Calculate n and phi
```

Name : Shubham Sanjay Karape

Roll No : T085

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```
n = p * q
phi = (p - 1) * (q - 1)

# Input e

e = int(input("Enter public exponent e :"))

if gcd(e, phi) != 1:
    print("Error: e must be coprime with phi.")
    exit()

# Find private key d

d = mod_inverse(e, phi)

if d is None:
    print("Error: Modular inverse not found.")
    exit()

# Input message

message = int(input("Enter numeric message (< {n}): "))

if message >= n:
    print("Error: Message must be smaller than n.")
    exit()

# Encryption

cipher = pow(message, e, n)
print("\nEncrypted Message:", cipher)

# Decryption

decrypted = pow(cipher, d, n)
print("Decrypted Message:", decrypted)

# Keys

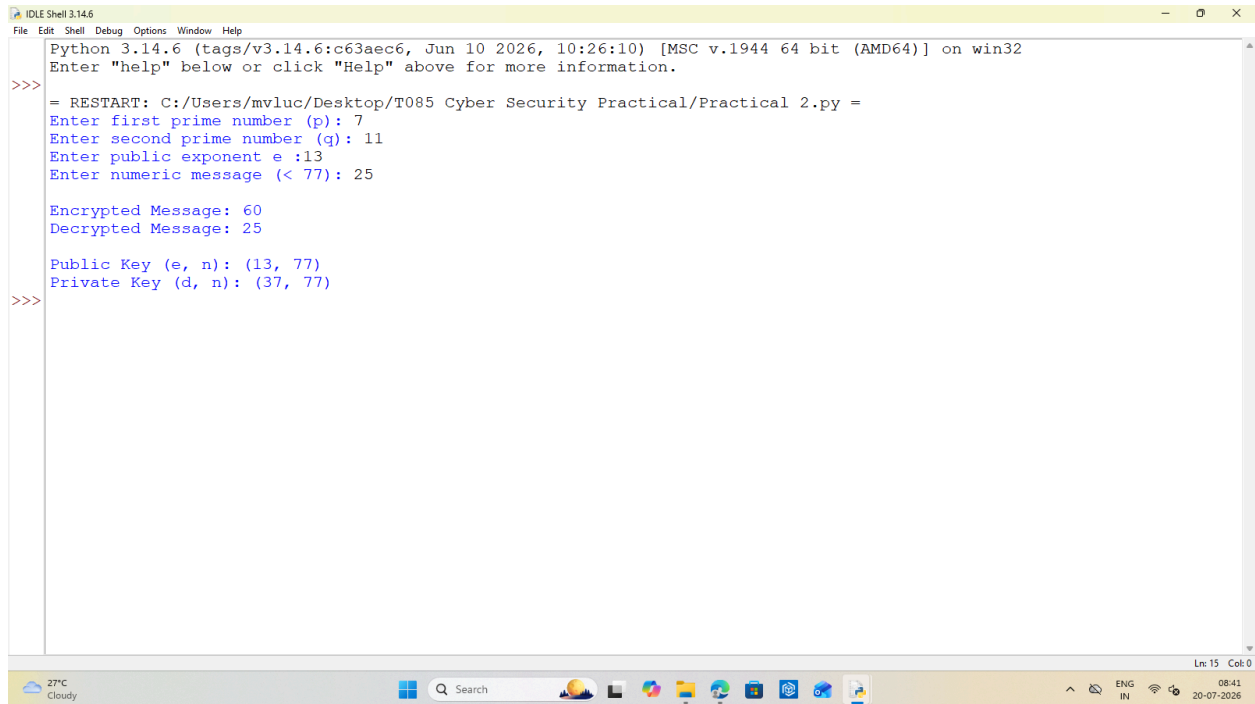
print("\nPublic Key (e, n):", (e, n))
print("Private Key (d, n):", (d, n))
```

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Output :



The screenshot shows a Python IDLE Shell window titled 'IDLE Shell 3.14.6'. The shell displays the following output:

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
= RESTART: C:/Users/mvluc/Desktop/T085 Cyber Security Practical/Practical 2.py =
Enter first prime number (p): 7
Enter second prime number (q): 11
Enter public exponent e :13
Enter numeric message (< 77): 25

Encrypted Message: 60
Decrypted Message: 25

Public Key (e, n): (13, 77)
Private Key (d, n): (37, 77)

>>>
```

The window's status bar at the bottom shows the system tray with weather (27°C Cloudy), search, taskbar icons, language (ENG IN), and date/time (08:41 20-07-2026).

GUI Version :

Code :

```
import tkinter as tk
from math import gcd

#RSA Functions

def is_prime(n):
    if n < 2:
        return False
    for i in range(2, int(n**0.5)+1):
        if n % i == 0:
            return False
    return True

def mod_inverse(e, phi):
    for d in range(1, phi):
        if (d * e) % phi == 1:
            return d
    return None
```

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Functions

```
def generate():
    global n, e, d

    p = int(ep.get())
    q = int(eq.get())
    e = int(ee.get())

    if not is_prime(p) or not is_prime(q):
        pub.config(text="Invalid Prime Numbers")
        return

    n = p * q
    phi = (p - 1) * (q - 1)

    if gcd(e, phi) != 1:
        pub.config(text="Invalid e")
        return

    d = mod_inverse(e, phi)

    pub.config(text=f"Public Key : ({e}, {n})")
    pri.config(text=f"Private Key : ({d}, {n})")

def encrypt():
    m = int(em.get())
    c = pow(m, e, n)
    cipher.config(text=f"Cipher : {c}")

def decrypt():
    c = int(ec.get())
    m = pow(c, d, n)
    plain.config(text=f"Message : {m}")

def clear():
    ep.delete(0, tk.END)
    eq.delete(0, tk.END)
    ee.delete(0, tk.END)
    em.delete(0, tk.END)
    ec.delete(0, tk.END)

    pub.config(text="")
    pri.config(text="")
    cipher.config(text="")
    plain.config(text="")
```

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#Window

```
root = tk.Tk()
root.title("RSA Algorithm")
root.state("zoomed")
root.configure(bg="black")
```

```
root.bind("<Escape>", lambda e: root.state("normal"))
```

```
frame = tk.Frame(root, bg="black")
frame.place(relx=0.5, rely=0.5, anchor="center")
```

```
title = tk.Label(
    frame,
    text="RSA ENCRYPTION",
    font=("Arial", 24, "bold"),
    bg="black",
    fg="cyan"
)
title.grid(row=0, column=0, columnspan=2, pady=20)
```

#Inputs

```
labels = ["Prime p", "Prime q", "Public e", "Message", "Cipher"]
```

```
entries = []
```

```
for i, text in enumerate(labels):
    tk.Label(frame,
        text=text,
        bg="black",
        fg="cyan",
        font=("Arial", 12)).grid(row=i+1, column=0, pady=8)
```

```
entry = tk.Entry(frame,
    width=25,
    bg="#111111",
    fg="cyan",
    insertbackground="cyan")
entry.grid(row=i+1, column=1, pady=8)
entries.append(entry)
```

```
ep, eq, ee, em, ec = entries
```

#Buttons

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```
tk.Button(frame, text="Generate Keys",  
          command=generate,  
          bg="cyan").grid(row=6, column=0, pady=15)
```

```
tk.Button(frame, text="Encrypt",  
          command=encrypt,  
          bg="cyan").grid(row=6, column=1)
```

```
tk.Button(frame, text="Decrypt",  
          command=decrypt,  
          bg="cyan").grid(row=7, column=0)
```

```
tk.Button(frame, text="Clear",  
          command=clear,  
          bg="cyan").grid(row=7, column=1)
```

#Outputs

```
pub = tk.Label(frame, text="", bg="black", fg="cyan", font=("Arial", 12))  
pub.grid(row=8, column=0, columnspan=2, pady=5)
```

```
pri = tk.Label(frame, text="", bg="black", fg="cyan", font=("Arial", 12))  
pri.grid(row=9, column=0, columnspan=2, pady=5)
```

```
cipher = tk.Label(frame, text="", bg="black", fg="cyan", font=("Arial", 12))  
cipher.grid(row=10, column=0, columnspan=2, pady=5)
```

```
plain = tk.Label(frame, text="", bg="black", fg="cyan", font=("Arial", 12))  
plain.grid(row=11, column=0, columnspan=2, pady=5)
```

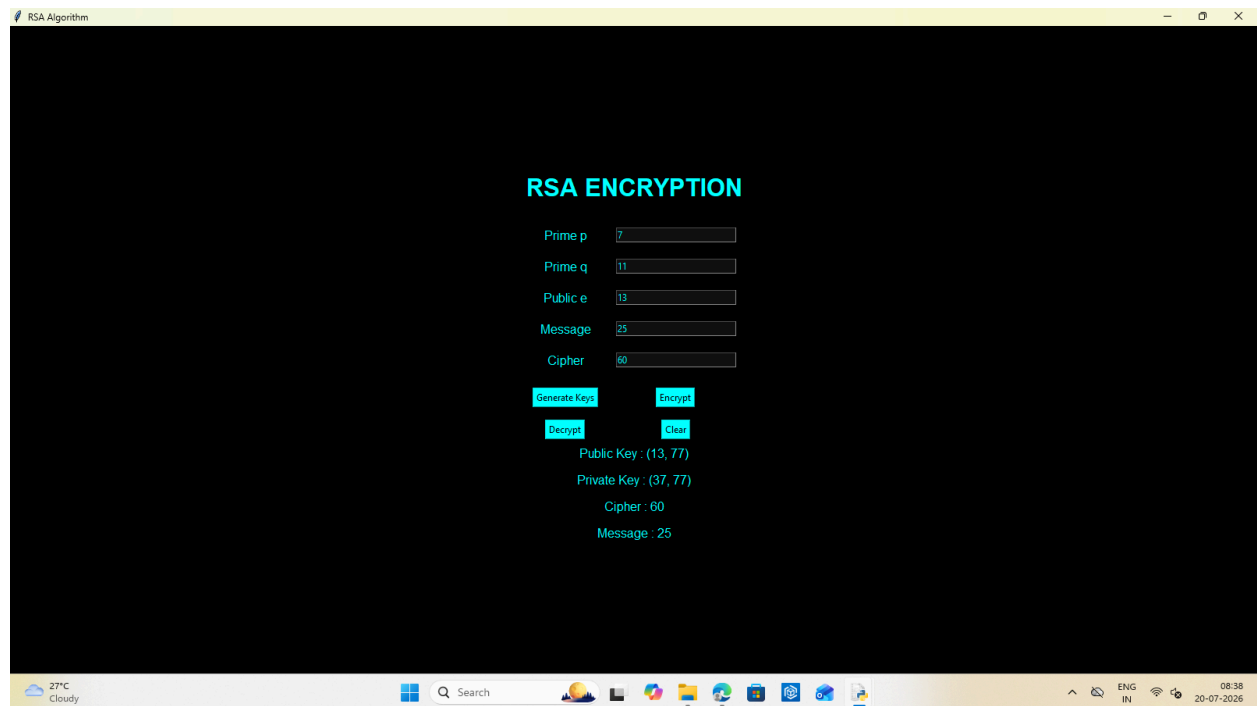
```
root.mainloop()
```

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Output :



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